Practical - 1

Practical Title: Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

Problem Definition 1: Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and GitHub repository.

A. Theory

1. What is URL and parts of URL?

Answer: URL means (Uniform Resource Locator) . It is the address of something on the internet. It have parts like:
1)Protocol: https or http
2)Domain name: Host's Name
3)Port Number: 443,200,etc
4)Path: Specific address to something in the internet.
I used the help of Google Gemini and ChatGpt.

2. How does HTML file process in web browser?

Answer:

User Opens URL / HTML File

Browser Sends Request

Server Returns HTML File

Browser Parses HTML

Creates DOM

Loads CSS & JavaScript


Renders and Displays Webpage

References: Duckett, J. (2021). *HTML and CSS: Design and Build Websites*. John Wiley & Sons.

3. How will page navigation flow managed among all HTML pages?
Answer: A sitemap is an important part of a website because it improves navigation for users and helps search engines understand the website's structure. This makes the website more user-friendly and improves its visibility in search engine results.

Uses of a Sitemap

1. **Improves Website Navigation**
    - A sitemap helps visitors quickly locate the information or page they are looking for.
2. **Helps Search Engines**
    - Search engines use sitemaps to find and index website pages, making them easier to appear in search results.
3. **Organizes Website Structure**
    - It provides a clear overview of all the pages and their relationships within the website.
4. **Speeds Up Indexing**
    - Newly added or updated pages can be discovered and indexed faster by search engines.
5. **Useful for Large Websites**
    - Websites with many pages benefit from sitemaps because they make content easier to manage and access.

References:

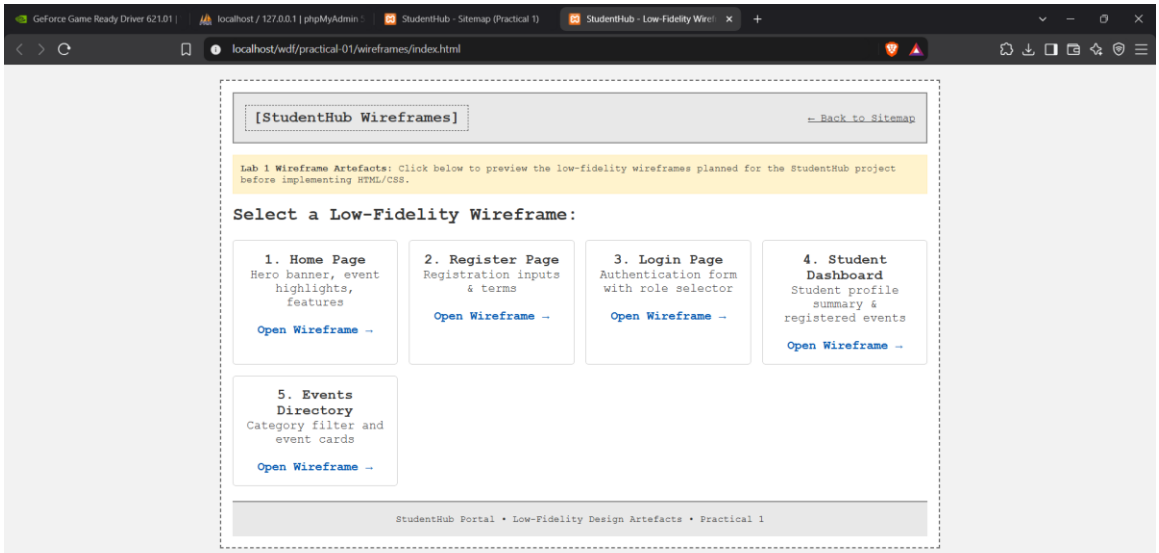
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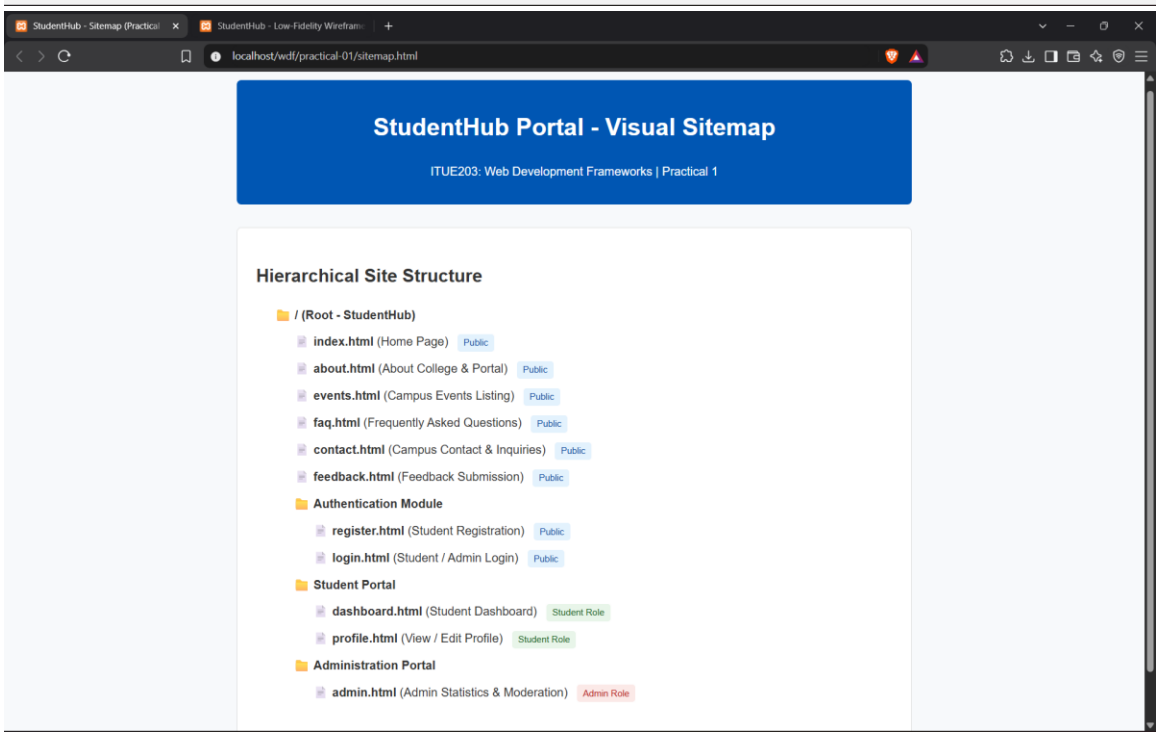
4. How will GitHub commits be maintained after each practical?
Answer: GitHub commits should be maintained by making a commit after completing each practical. Every commit should include only the changes made for that particular practical and have a clear, meaningful commit message describing what was done. This creates a record of the work completed over time and makes it easy to track progress.

Example Commit Messages

- Added Practical 1 - HTML Basics
- Completed Practical 2 - Lists and Tables
- Updated Practical 3 - Forms
- Fixed errors in Practical 4

B. Evidence of Work (Screenshots):





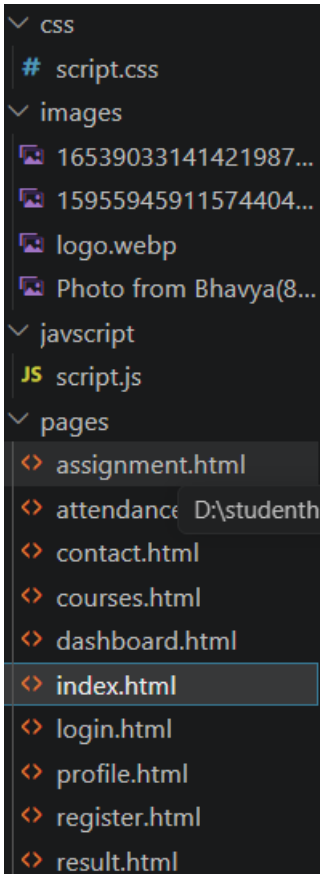
GitHub Repository Link

https://github.com/dhyandesai1727/25DIT012-WDF-BATCH-A

C. Challenges Faced:

GITHUB REPO. CREATION,FOLDER CREATION,FILE CREATION.

VSCODE FOLDER,FILE CREATION.



D. Learning Outcomes:

I LEARNED WEBSITE CREATION, SITEMAP, WIREFRAME, GITHUB, REQUIREMENT ANALYSIS, FILE MANAGEMENT